

1. Alice and Song

Alice has a collection of songs represented as a string S where each character represents a song. A playlist is a substring of the given string with exactly K number of songs. She wants to create a playlist that contains maximum number of her favourite song which is 'a'. Your task is to find and return an integer value representing the maximum number of favourite songs that she can get in a single playlist.

Input Specification:

input1 : A string value S representing the collection of songs.

input2 : An integer value K representing the number of songs in the playlist.

Output Specification: Return an integer value representing the maximum number of favourite songs that she can get in a single playlist.

Example 1:

input1: abaca

input2: 3

Output: 2

Explanation:

Here, the given collection of songs is "abaca" and K is 3. Now, all the possible playlists of K songs are "aba", "bac", "aca". In each playlist maximum of two a's were there so the output is 2

Example 2:

input1 : bcdefgfedcb

input2 : 5

Output : 0

Explanation:

Here, the given collection of songs is "bcdefgfedcb" and K is 5. As we can see, there is no favourite song i.e 'a'. Therefore, 0 is returned as the output.

Solution

```
def maxFavSong(S, K):  
    m = 0  
    c = 0  
    for i in range(K):  
        if S[i] == 'a':  
            c += 1  
    m = c  
    for i in range(K, len(S)):  
        if S[i - K] == 'a':  
            c -= 1  
        if S[i] == 'a':  
            c += 1  
        m = max(m, c)  
    return m  
  
str = input()  
n = int(input())  
print(maxFavSong(str, n))
```

2. John is given a binary array of N integers and he needs to convert all the integers to either 0 or 1, The cost of changing from 0 to 1 is X coins and the cost of changing from 1 to 0 is Y coins. Your task is to find and return an integer value representing the minimum amount of coins needed to convert the array as desired.

Input Specification:

input1 : An integer value N, representing the length of the binary array

input2 : An integer value X, representing the cost of converting 0 to 1

input3 : An integer value Y, representing the cost of converting 1 to 0

input4 : An integer array of length N.

Output Specification:

Return an integer value representing the minimum amount of coins needed to convert the array as desired.

Example 1:

input1 : 4

input2 : 5

input3 : 4

input4: [1, 1, 0, 0]

Output: 8

Explanation: To convert all elements in the array [1, 1, 0, 0] to 0s, the cost is (2 times 4 = 8) coins. To convert all elements to 1s, the cost is (2 times 5 = 10) coins. The minimum cost is 8 coins, so the output is 8.

Solution

```
def minConvCost(N, X, Y, arr):  
    cz = 0  
    c1 = 0  
    for num in arr:  
        if num == 1:  
            cz += Y  
        else:  
            c1 += X  
    return min(cz, c1)  
N = int(input())  
X = int(input())  
Y = int(input())  
arr = list(map(int, input().split()))  
print(minConvCost(N, X, Y, arr))
```

3. Bob and roof

Bob has a pair of magical shoes that allows him to climb 3 stairs in one go. There are N houses whose roofs Bob wants to reach. The number of stairs to the roof of each house is given in an array A . Bob can reach only the roofs of those houses whose stairs are multiples of 3 but he has the additional power of removing stairs from adding stairs to any house. Your task is to find and return the sum of the minimum number of stairs needed to be added to or removed from each house so that Bob can climb all the roofs.

Input Specification:

input1 : An integer value N representing the number of houses.

input2 : An integer array A representing the number of stairs in each house.

Output Specification: Return an integer value representing the sum of the minimum number of stairs needed to be added to or removed from each house so that Bob can climb all the roofs

Solution

```
def minStairs(N, A):  
    sum = 0  
    for s in A:  
        rem = s % 3  
        if rem == 1:  
            sum += 1  
        elif rem == 2:  
            sum += 1  
    return sum
```

```
N = int(input())  
A = list(map(int, input().split()))  
print(minStairs(N, A))
```

Example 1:

input1 : 4

input2 : {12,21,3,4}

Output : 1

Explanation: Here, the given array is {12,21,3,4}. Bob will not be able to climb to the roof of the 4th house as the number of stairs is not a multiple of 3. He can do either of the following:

1. Add 2 stairs and make it 6.
2. Remove 1 stair and make it 3.

As we need to choose the option with the addition or removal of the minimum number of stairs, so Bob will choose the 2nd option. Therefore, 1 is returned as the output.

4. A public library has a simple policy for late returns i.e. a fixed fee Z is charged for each day X , a book is overdue but there's a grace period represented by an integer Y allowing returns within a certain number of days past the due date without incurring a fee. However, the policy includes a cap represented by an integer A on the total fee beyond which no additional charges are applied, regardless of how late the book is returned. Your task is to find and return an integer value representing the total late fee for a returned book

Input Specification:

input1 : An integer value X representing the number of days a book has been returned late.

input2 : An integer value Y representing the number of days past the due date that can elapse without incurring a fee.

input3 : An integer value Z representing the fixed fee charged for each day the book is overdue beyond the grace period.

input4 : An integer value A representing the maximum total fee that can be charged.

Output Specification: Return an integer value representing the total late fee for a returned book.

Example-1:

Input-1: 12

Input-2: 5

Input-3: 3

Input-4: 20

Output: 20

Explanation:

- The book was returned 12 days late.
- The library allows a grace period of 5 days, so no fee is charged for the first 5 days.
- After the grace period, the book is 7 days late.
- The fixed fee is 3 per day, so for 7 days, the fee would be (7 times 3 = 21).
- However, the maximum fee that can be charged is 20, so the total fee is capped at 20.

Solution

```
def calculate_late_fee(X, Y, Z, A):  
    days_beyond_grace = max(0, X - Y)  
    total_fee = days_beyond_grace * Z  
    return min(total_fee, A)
```

```
X = int(input())  
Y = int(input())  
Z = int(input())  
A = int(input())  
print(calculate_late_fee(X, Y, Z, A))
```

5. Car's speed

You have an electric car whose speed is derived from the battery percentage. The battery percentage is denoted by an integer value N and the speed is the sum of all the factors of the battery percentage including 1. Your task is to find and return an integer value representing the speed of the car.

Input Specification:

input1 : An integer value N representing the battery percentage of the car.

Output Specification:

Return an integer value representing the speed of the car

Example 1:

input1 : 12

Output : 28

Explanation:

Here, the given battery percent is 12 and the factors of 12 are 1, 2, 3, 4, 6, and 12. The sum of all the factors will be $1 + 2 + 3 + 4 + 6 + 12 = 28$. Therefore 28 is returned as the output.

Solution

```
def car_speed(N):  
    sum1=0  
    i=1  
    while(i<=N/2):  
        if(N%i==0):  
            sum1+=i  
        i+=1  
    sum1+=N  
    return sum1  
N=int(input())  
print(car_speed(N))
```

6. Chocolate Jar

You are given an integer array of size N, representing jars of chocolates. Three students A, B, and C respectively, will pick chocolates one by one from each chocolate jar, till the jar is empty, and then repeat the same with the rest of the jars. Your task is to find and return an integer value representing the total number of chocolates that student A will have, after all the chocolates have been picked from all the jars.

Input Specification:

input1 : An integer array representing the quantity of chocolates in each jar

input2 : An integer value N representing the number of jars.

Output Specification:

Return an integer value representing the total number of chocolates that student A will have, after all the chocolates are picked.

Example1:

input1: {3,2,6}

input2: 3

Output: 4

Solution

```
def total_chocolates_for_A(jars, N):  
    A_total = 0  
    for i in range(N):  
        chocolates_in_jar = jars[i]  
        A_total += chocolates_in_jar // 3  
        if chocolates_in_jar % 3 > 0:  
            A_total += 1  
    return A_total  
  
input1 = list(map(int, input().split()))  
input2 = int(input())  
print(total_chocolates_for_A(input1, input2))
```

7.Chocolate Types

Monica went to a confectionery shop which has N different types of chocolates. The shop displays a price chart, where column T represents the types of chocolate and column P represents their respective prices. She was asked to pick K different types of chocolates at the minimum possible cost. Your task is to find and return an integer value representing the total amount of money she will spend. Return -1, if K types of chocolates cannot be chosen.

Note: She can buy only 1 chocolate of each type.

Input Specification:

input1 : An integer value N , representing the number of chocolate types

input2 : An integer value K , representing the number of types of chocolates to buy

input3 : An integer array representing the types of chocolates T

input4 : An integer array representing the prices P

Output Specification:

Return an integer value representing the total money she will spend.

Return -1, if K types of chocolates cannot be chosen

Example 1:

input1: 3

input2 : 1

input3 : (1,2,3)

input4 : (2,1,3)

Output : 1

Explanation: Here, the given array for the types of chocolate is {1,2,3} and the respective price array is {2,1,3}. As the value of K is 1. so Monica can buy only 1 type of chocolate.

The chocolate costing the minimum among all the different categories is the second chocolate with price 1. The total money spent by Monica will be 1, so 1 is returned as the output.

Example 2:

input1 : 8

input2 : 3

input3 : {1,3,2,2,4,1,3,5}

input4 : {3,3,0,1,2,4,1,4}

Output : 3

Explanation: Here, the given array for the types of chocolate is {1,3,2,2,4,1,3,5} and the respective price array is {3,3,0,1,2,4,1,4}. As the value of K is 3, Monica can buy 3 types of chocolates. The chocolates costing the least among all the different categories are : 1. Type 2 chocolate with price 0. 2. Type 4 chocolate with price 2. 3. Type 3 chocolate with price 1. The total money spent by Monica will be 3 (0 + 2 + 1), so 3 is returned as the output.

Solution

```
def minCost(N, K, T, P):  
    if K > N:  
        return -1  
    P.sort()  
    total_cost = sum(P[:K])  
    return total_cost
```

```
N = int(input())  
K = int(input())  
T = list(map(int, input().split()))  
P = list(map(int, input().split()))  
print(minCost(N, K, T, P))
```

8. Circuit Board

There is a board with N slots. Each slot contains a single coin with integer numbers printed on it. The final sum of the entire board is calculated as $A_1 - A_2 + A_3 - A_4 + A_5 \dots A_N$. You can shuffle the coins in each block any number of times. Your task is to find and return the maximum sum after the optimal arrangement.

Input Specification:

input1 : An integer value N , representing the number of coins.

input2 : An integer array A , representing the numbers printed on the coins.

Output Specification:

Return an integer value representing the maximum sum after the optimal arrangement.

Example 1:

input1 : 8

input2 : {1,-6,-5,-12,1,3,8,-3}

Output : 39

Explanation:

Here, the given array is {1,-6,-5,-12,1,3,8,-3}, and one of the best arrangements which will give the maximum sum will be {8,-12,3,-6,1,-5,1,-3}. This will give the maximum sum value of 39. 39 is returned as the output.

Example 2:

input1: 5

input2 : {-99,-1,0,50,10}

Output : 160

Explanation:

Here, the given array is {-99,-1,0,50,10}, and one of the best arrangements which will give the maximum sum will be {50,99,10,-1,0}. This will give the maximum Sum value of 160. Therefore, 160 is returned as the output.

Solution

```
def max_sum(N, A):  
    A.sort()  
    max_sum = 0  
    for i in range(N):  
        if i % 2 == 0:  
            max_sum += A[-(i // 2 + 1)]  
        else:  
            max_sum -= A[i // 2]  
  
    return max_sum  
  
N = int(input())  
A = list(map(int, input().split()))  
print(max_sum(N, A))
```

9. Clean String

You are given two strings A and B. Your task is to find and return a string representing the leftover string in A after removing all the letters that exist in string B. Return "Empty" if the output does not contain any value.

Note : Strings A and B contain English alphabets in upper case only.

A single alphabet in B can replace all the occurrences of that alphabet in A.

Input Specification:

input1 : A string A

input2 : A string B

Output Specification:

Return a string representing the leftover string in A after removing all the letters that exist in string B.

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Example 1:**input1** : AABBBCC**input2** : AB**Output** : CC

Explanation: Here, the given string A is "AABBBCC" and B is "AB". We need to remove all the occurrences of characters 'A' and 'B' from string A. After removing them, we will be left with only 'CC'. Therefore, CC is returned as the output.

Example 2:**input1** : AABBBCC**input2** : ABCDDSD**Output** : Empty**Explanation:**

Here, the given string A is "AABBBCC" and B is "ABCDDSD". We need to remove all the occurrences of characters 'A', 'B', 'C', 'D' and 'S' from string A. After removing them, we will be left with an empty string, so

Empty is returned as the output.

Solution



```
def clean_string(A, B):  
    remove_set = set(B)  
    result = [char for char in A if char not in remove_set]  
    cleaned_string = "".join(result)  
    if not cleaned_string:  
        return "Empty"  
  
    return cleaned_string  
  
A = input()  
B = input()  
print(clean_string(A, B))
```

10. Consecutive Difference

There is a coding project which takes an array A of integers. Your task is to calculate the difference between each consecutive pair of numbers and return an integer array representing the differences of consecutive numbers.

Note: The difference should be calculated as the second number minus the first number.

Input Specification:

input1 : An integer array A containing numbers.

Output Specification:

Return an integer array representing the differences of consecutive numbers.

Example 1:

input1 : {5,2,10,8,1}

Output : {-3,8,-2,-7}

Explanation:

Here, A = {5,2,10,8,1}. The difference between the first pair (5 and 2) is -3. The difference between the second pair (10 and 2) is 8 and so on. Therefore, {-3,8,-2,-7} will be returned as output.

Example 2:

input1 : {20,15,10,5,0}

Output : {-5,-5,-5,-5}

Explanation:

Here, A = {20,15,10,5,0}. The difference between the first pair (20 and 15) is -5. The difference between the second pair (10 and 15) is -5 and so on. Therefore, {-5,-5,-5,-5} will be returned as output.

```
def consecutive_diff(A):  
    diffs = [A[i + 1] - A[i] for i in range(len(A) - 1)]  
    return diffs
```

```
A = list(map(int, input().split()))  
print(consecutive_diff(A))
```

11. Cupcake Sales

You are given an array A of size N , where each element represents the number of cupcakes sold in a single transaction. Your task is to find and return an integer value representing the sum of the cupcakes from the transactions where 5 or more cupcakes were sold. Return 0 if there is no transaction with more than 5 cupcakes sold.

Input Specification:

input1 : An integer value N representing the number of transactions.

input2 : An integer array A representing the number of cupcakes sold in each transaction.

Output Specification:

Return an integer value representing the sum of the cupcakes from the transactions where 5 or more cupcakes were sold. Return 0 if there is no transaction with more than 5 cupcakes sold.

Example 1:

input1:5

input2 :{3,5,17,2,9}

Output : 31

Explanation:Here, the given array is {3,5,17,2,9}.

The sum of cupcakes from the transactions where 5 or more cupcakes were sold is 31 (5+17+9). Therefore, 31 is returned as the output.

Example 2:

input1 : 4

input2 : {1,4,3,3}

Output : 0

Explanation:Here, the given array {1,4,3,3}

Since there is no transaction with more Than 5 cupcakes sold. Therefore 0 is returned as the output.

Solution

```
def cupcake_sales(N, A):  
    total = 0  
    for cupcakes in A:  
        if cupcakes >= 5:  
            total += cupcakes  
    return total  
  
N1 = int(input())  
A1 = list(map(int, input().split()))  
print(cupcake_sales(N1, A1))
```

12. Dinner Dishes



You are planning a dinner menu for an event, and you have a list of N dishes stored in array A , each with a certain cost associated with it. You must select two such dishes, such that the sum of their costs is maximum among all available pairs. Your task is to find and return an integer value representing the sum of such available pair. Return 0 in case of no such pair available.

Input Specification:

input1 : An integer array A representing the cost of dishes.

input2 : An integer value N representing the number of dishes available.

Output Specification:

Return an integer value representing sum of such pair available. In case, there are no such pair present, return 0.

Example 1:

input1 : {1,10,5,15}

input2 : 4

Output : 25

Explanation:

Here, $A = \{1, 10, 5, 15\}$. The pair with maximum sum would be 10 and 15, since they are the largest in the array of prices and their sum would be 25. Therefore, 25 is returned as the answer.

Example 2:

input1 : {50,50,50}

input2 : 3

Output : 100

Explanation:

Here, $A = \{50, 50, 50\}$. The pair with maximum sum would be 50 and 50, since they are the largest in the array of prices and their sum would be 100. Therefore, 100 is returned as the answer.

Solution

```
def max_dish_pair_sum(A, N):  
    if N < 2:  
        return 0  
    A.sort(reverse=True)  
    return A[0] + A[1]
```

```
A = list(map(int, input().split()))  
N = int(input())  
print(max_dish_pair_sum(A, N))
```

13. Divide String

You are given a string S. You have to divide the string into subsequences such that every character of S is present in only one subsequence and every subsequence has the same set of characters. Your task is to find and return the maximum possible number of subsequences into which you can divide the given string.

Input Specification:

input1 : A string value S.

Output Specification:

Return an integer value representing the maximum possible number of subsequences into which you can divide the given string.

Example 1:

input1 : abcabcab

Output : 2

Explanation:

Here, the given string is "abcbcab", We can divide the string into two subsequences "abca", and "bcab".

Both have the same sets of characters, i.e., ['a', 'b', 'c']. Therefore, 2 is returned as the output.

Solution

```
def max_subsequences(S):  
    freq = {}  
    for char in S:  
        if char in freq:  
            freq[char] += 1  
        else:  
            freq[char] = 1  
    min_freq = min(freq.values())  
    return min_freq
```

```
S1 = input()  
print(max_subsequences(S1))
```

Solution

```
def max_subsequences(S):  
    freq = {}  
    for char in S:  
        if char in freq:  
            freq[char] += 1  
        else:  
            freq[char] = 1  
    min_freq = min(freq.values())  
    return min_freq
```

```
S1 = input()  
print(max_subsequences(S1))
```

Solution

```
def find_majority_party(n, votes):  
    if n <= 1:  
        return -1  
    vote_count = {}  
    for vote in votes:  
        if vote in vote_count:  
            vote_count[vote] += 1  
        else:  
            vote_count[vote] = 1  
    majority_needed = n / 2  
    for party, count in vote_count.items():  
        if count > majority_needed:  
            return party  
    return -1
```

```
N = int(input())  
A = list(map(int, input().split()))  
print(find_majority_party(N, A))
```

Example 1:**input1:**7**input2 :** {1,2,2,2,3,3,2}**Output :** 2**Explanation:**

Here, the party with the number 2 has the maximum number of votes, which is greater than half of the votes cast in the election. Therefore, 2 is returned as the output.

Example 2:**input1 :** 3**input2 :** {1,2,3}**Output :** -1

Explanation: Here, all the three parties have the same number of votes cast for them.

15. Even Sum

You are organizing a charity run where participants contribute a dollar for every even-numbered kilometer they complete. Your task is to find and return an integer value representing the total amount of money raised if the race is N km long.

Input Specification:

input1 : An integer value N

Output Specification:

Return the sum of all even numbered kilometers till N they complete.

Example 1:

input1 : 10

Output : 30

Explanation:

Here, $N = 10$. The even numbers between 1 and 10 are 2, 4, 6, 8, 10 and their sum will be 30. Hence, 30 is returned as the output.

Example 2:

input1 : 7

Output : 12

Explanation:

Here, $N = 7$. The even numbers between 1 and 7 are 2, 4, 6 and their sum will be 12. Hence, 12 is returned as the output.

Solution

```
def evenSum(N):  
    sum=0  
    for i in range(1,N+1):  
        if i%2==0:  
            sum=sum+i  
    return sum
```

```
N=int(input())  
retVal=evenSum(N)  
print(retVal)
```

16. Financial Dataset

You are working on a financial analyzing tool which represents the daily stock process of a company over a time. Each element in an integer array A of size N represents the closing price of the stock for a particular day. Your task is to find and return an integer value representing the total number of days where the stock price decreased indicating negative growth.

Input Specification:

input1 : An integer array A containing the closing price of the stock.

input2 : An integer value N representing the size of array

Output Specification:

Return an integer value representing the total number of days where the stock price decreased, indicating negative growth.

Example 1:

input1 : {3,5,6,-7,9,10,-12}

input2 : 7

Output : 2

Explanation:

Here, $A = \{3, 5, 6, -7, 9, 10, -12\}$. The stock price decreased at -7 and -12. Therefore, 2 will be returned as output.

Example 2:

input1 : {-5,-8,-12}

input2 : 3

Output : 3

Solution

```
def stockPriceDecrease(arr,n):  
    count = 0  
    for i in range(n):  
        if arr[i]<1:  
            count+=1  
    return count
```

```
n=int(input())  
arr=list(map(int,input().split()))[:n]  
retVal=stockPriceDecrease(arr,n)  
print(retVal)
```

17. Find Letters

Your friend gives you a string S which contains the name of a person and asks you to find the first and last letter of their name. Your task is to find and return a string representing the first and the last letter.

Note:

The output should be two capital letters with a dot separating them.

The output is case sensitive.

Input Specification:

input1 : A string S representing the name

Output Specification:

Return a string representing the first and the last letter.

Solution

```
def firstLastLetter(str1):  
    result=str1[0].upper()+ "." +str1[-1].upper()  
    return result
```

```
str1=input()  
retVal=firstLastLetter(str1)  
print(retVal)
```

Example 1:

input1 : Sam Harris

Output : S.S

Explanation:

Here, the name is 'Sam Harris'. The first and last letters of this name are 'S' and 'S'.

Hence, S.S is returned as the output.

Example 2:

input1 : Patrick Feeney

Output : P.Y

Explanation:

Here, the name is 'Patrick Feeney'. The first and last letters of this name are 'P' and 'Y'. Hence, P.Y is returned as the output.

18. Fitness App

You are developing a small application for a fitness app. In the app, users usually enter their workout duration in N minutes. Your task is to convert the minutes and return an integer representing the equivalent value in seconds.

Input Specification:

input1 : An integer value N representing the workout in minutes.

Output Specification:

Return an integer representing the equivalent value of N in seconds.

Example 1:

input1: 5

Output : 300

Explanation:

Here, $N = 5$. The conversion of 5 minutes will be $5 \times 60 = 300$ seconds. Hence, 300 is returned as output.

Example 2:

input1 : 3

Output : 180

Explanation:

Here, $N = 3$. The conversion of 3 minutes will be $3 \times 60 = 180$ seconds. Hence, 180 is returned as output

Solution



```
def fitnessTime(tm):  
    ts=tm*60  
    return ts
```

```
tm=int(input())  
retVal=fitnessTime(tm)  
print(retVal)
```

19. Goal Tracker



You are developing a feature for a health and fitness app that helps users track their daily steps. Users input the total number of daily steps S , the number of steps they have completed D , and the number of days they have been tracking their steps N . Your task is to find and return an integer value representing the average number of steps the user needs to take per day to reach their target.

Note: Round off the output to its nearest value.

Input Specification:

input1 : An integer value S representing the total number of daily steps.

input2 : An integer value D representing the number of steps user has completed.

input3 : An integer value N representing the number of days the user has been tracking their steps.

Output Specification:

Return an integer value representing the average number of steps the user needs to take per day to reach their target.

Example 1:

input1:348

input2:327

input 3: 5

Output : 4

Explanation:

Here, $S=348$, $D=327$ and $N=5$. The user needs to take 21 more steps ($348-327=21$). Since they have been tracking their steps for 5 days, the average steps per day needed to reach the target is $21/5$, which is approximately 4. Hence, 4 is returned as output.

Example 2:

input1 : 6000

input2 : 2455

input3 : 7

Output : 506

Explanation:

Here, $S=6000$, $D=2455$ and $N=7$. The user needs to take 3545 more steps ($6000-2455$). Since they have been tracking their steps for 7 days, the average steps per day needed to reach the target is $3545/7$, which is approximately 506. Hence, 506 is returned as output.

Solution



```
def averageSteps(S,D,N):  
    steps_pending=S-D  
    avg_steps=steps_pending // N  
    return avg_steps
```

```
S=int(input())  
D=int(input())  
N=int(input())  
retVal=averageSteps(S,D,N)  
print(retVal)
```

Solution

```
def lastImportantSection(sections):  
    for i in range(len(sections)-1,-1,-1):  
        if sections[i]=='1':  
            return i  
    return -1
```

```
str1 = "00010000"  
print(lastImportantSection(str1))
```

20. Important Section

In a digital document, you are marking certain sections as important using the character '1', while all the other sections are marked with 0. Your task is to find and return an integer value representing the last marked important section. Return -1, in case of no important sections.

Input Specification:

input1 : A string value representing the sections

Output Specification:

Return an integer value representing the last marked important section.

Example 1:

input1 : 00010000

output: 3

21. Magic String

Eva has a string S containing lowercase English letters. She wants to transform this string into a Magic String, where all the characters in the string are the same. To do so, she can replace any letter in the string with another letter present in that string.

Your task is to help Eva find and return an integer value, representing the minimum number of steps required to form a Magic String. Return 0, if S is already a Magic String.

Input Specification:

input1 : A string S , containing lowercase English letters.

Output Specification:

Return an integer value, representing the minimum number of steps required to form a Magic String.
Return 0, if S is already a Magic String.

Example 2:

input1 : aaaa

Output : 0

Explanation:

Here in this example, string S is already a Magic string because all its letters are the same.

Hence, 0 is returned as the output.

Example 1:

input1 : aaabbbccdddd

Output : 8

Explanation:

Here in this example, the string S is aaabbbccdddd. We need perform the following operations to make

Magic String:

1st way:

On replacing the letters b, c, d with a, the Magic string formed is aaaaaaaaaaaaaa.

Here, the minimum number of steps to convert the original string to Magic String is 9.

Optimal Way:

On replacing the letters, a, b, c with d, the Magic string formed is dddddddddddd.

Here, the minimum number of steps required to form the Magic String is 8.

Since 8 is smaller than 9, 8 is returned as the output.

Solution

```
def magicString(str1):  
    max=0  
    for i in str1:  
        if str1 == str1.replace(str1, i):  
            return 0  
        c=str1.count(i)  
        if c>max:  
            max=c  
    return len(str1)-max  
  
str1=input()  
print(magicString(str1))
```

22. Mathematics Tutoring



You are developing a small application for a mathematics tutoring center. The center offers online resources to help students with their math homework. As a part of it, you are assigned with a task to develop an application in which, if a student enters a number N , then it calculates and returns an integer which is the factorial of the entered number. Your task is to find and return the same number.

Input Specification:

input1 : An integer value N representing the number entered by student.

Output Specification:

Return an integer value representing the factorial of the entered number

Example 1:

input1 : 5

Output : 120

Explanation:

Here, $N = 5$. The factorial of 5 will be $(5 \times 4 \times 3 \times 2 \times 1 = 120)$ 120. Hence, 120 is returned as output.

Example 2:

input1: 3

Output:6

Explanation:

Here, $N=3$. The factorial of 5 will be $(3 \times 2 \times 1 = 6)$ 6. Hence, 6 is returned as output.

Solution

```
def factorial(N):  
    fact=1  
    for i in range(1,N+1):  
        fact=fact*i  
    return fact
```

```
N=int(input())  
print(factorial(N))
```

Example 1:

input1 :4

Output : 5

Explanation:

Here, the given value of $N = 4$, and the smallest prime number greater than 4 is 5. Therefore, 5 is returned as the output.

Example 2:

input1 : 12

Output : 13

Explanation:

Here, the given value of $N = 12$, and the smallest prime number greater than 12 is 13. Therefore, 13 is returned as the output.

Solution

```
def isPrime(num):  
    if num < 2:  
        return False  
    for i in range(2, (num//2)+1):  
        if num % i == 0:  
            return False  
    return True
```

```
def smallest_prime(N):  
    largeN = N + 1  
    while True:  
        if isPrime(largeN):  
            return largeN  
        largeN += 1
```

```
N=int(input())  
print(smallest_prime(N))
```

24. Minimum Array Sum

Paul is given an array A of length N. He must perform the following operations on the array sequentially:

Choose any two integers from the array and calculate their average.

If an element is less than the average, update it to 0. However, if the element is greater than or equal to the average, he need not update it.

Your task is to help Paul find and return an integer value, representing the minimum possible sum of all the elements in the array by performing the above operations.

Note: An exact average should be calculated, even if it results in a decimal.

Input Specification:

input1 : An integer value N, representing the size of the array A

input2 : An integer array A.

Output Specification:

Return an integer value, representing the minimum possible sum of all the elements in the array by performing the above operations.

Example 1:

input1 : 4

input2 : 4,5,6,7

Output : 7

Explanation:

Calculate the average of any two elements, e.g., $((4 + 5)/2 = 4.5)$.

1. Update elements less than 4.5 to 0. So, 4 becomes 0.
2. Repeat the process for other pairs:
 - $((5+6)/2 = 5.5)$, update elements less than 5.5 to 0. So, 5 becomes 0.
 - $((6+7)/2 = 6.5)$, update elements less than 6.5 to 0. So, 6 becomes 0.
3. The final array is $([0, 0, 0, 7])$.
4. The minimum possible sum is $(0 + 0 + 0 + 7 = 7)$.

Solution

```
def min_array_sum(N, A):  
    A.sort()  
    for i in range(N):  
        for j in range(i + 1, N):  
            avg = (A[i] + A[j]) / 2  
            if A[i] < avg:  
                A[i] = 0  
            if A[j] < avg:  
                A[j] = 0  
    return sum(A)
```

```
N = int(input())  
A = list(map(int, input().split()))  
print(min_array_sum(N, A))
```

25. Minimum Unit

Tom runs a juice shop where he has two machines, M and N, for making juice. M is a big machine which can make juice for 100 customers in one go, while machine N can make juice for only 4 customers at a time.

M consumes X units of power, while N consumes Y units of power. Your task is to find and return an integer value representing the minimum number of units of power Tom's machines will consume to serve P customers. He can run any combination of M and N.

Input Specification:

input1 : An integer value P representing the number of customers

input2 : An integer value X representing the power consumption for M

input3 : An integer value Y representing the power consumption for N

Output Specification:

Return an integer value representing the minimum units of power Tom will use to serve P customers, while running any combination of M and N.

Example 1:**input1:** 9**input2 :** 40**input3 :** 8**Output :** 24**Explanation:**

Here, $P = 9$, $X = 40$ and $Y = 8$. Given below are the two approaches we can use:

If we use M, he can serve 9 customers in one go and this will consume 40 units of power.

If we use N, he can serve 12 customers in 3 cycles, but he needs to serve only 9 customers. So N needs to be run 3 times, which will consume $3 \times 8 = 24$ units.

Since 24 units is lesser than 40 units, it is better to use N 3 times, Therefore, 24 is returned as the output

Example 2:

input1 : 105

input2 : 80

input3 : 10

Output : 100

Explanation:

Here, $P = 105$, $X = 80$ and $Y = 10$. Given below are the approaches we can use.

If we run M twice, we can serve 200 customers, which is greater than 105, and this will consume 160 units of power.

If we run N 27 times, we will be able to serve 108 customers, which is greater than 105 and this will consume 270 units of power.

Optimally, if we run M once and N twice, this will consume a total power NEED + $2 \times 10 = 100$ units. Of all the above mentioned cases. Case 3 is the best, as it consumes the least number of units, viz., 100 units, Therefore, 100 is returned as the output

Solution

log

```
def min_power(P, X, Y):
    cap_M = 100
    cap_N = 4
    min_pow = float('inf')
    num_M = (P + cap_M - 1) // cap_M
    min_pow = min(min_pow, num_M * X)
    num_N = (P + cap_N - 1) // cap_N
    min_pow = min(min_pow, num_N * Y)
    for m in range(P // cap_M + 1):
        served_by_M = m * cap_M
        rem = P - served_by_M
        pow = m * X
        if rem > 0:
            num_N = (rem + cap_N - 1) // cap_N
            pow += num_N * Y
        min_pow = min(min_pow, pow)
    return min_pow

P = int(input())
X = int(input())
Y = int(input())
print(min_power(P, X, Y))
```

26. Operational Substring

You are given a string S. You want to make the maximum possible number of substrings of S with the same characters and for that you can swap any two characters of the string any number of times. Your task is to find and return an integer value representing the maximum possible number of substrings with same characters.

Note: Count every character as unique.

Input Specification:

input1: A string value S.

Output Specification:

Return an integer value representing the maximum possible number of substrings with same characters

Example 1:**input1 :** asda**Output :** 5**Explanation:**

Here, the given string is "asda". We can swap the characters of string S at index 1 and 3, making the string "aads". Now there are 5 possible substrings, ("a", "aa", "a", "d", "s"), which have the same characters. Therefore, 5 is returned as the output

Example 2:**input1 :** xyxy**Output :** 6**Explanation:**

Here, the given string is "xyxy". We can swap the characters of string S at index 1 and 2, making the string "xxyy". Now there are 6 possible substrings, {"x" xx", "x", 'y', "yy", "y"}, which have the same characters. Therefore, 6 is returned as the output.

Solution

```
def max_same_char_substrings(S):  
    freq = {}  
    for char in S:  
        if char in freq:  
            freq[char] += 1  
        else:  
            freq[char] = 1  
    max_substrings = 0  
    for count in freq.values():  
        max_substrings += count * (count + 1) // 2  
    return max_substrings  
S1 = input()  
print(max_same_char_substrings(S1))
```

27. Prime Number Picnic

You are planning a picnic for a group of friends who love math. To make it interesting, you decided to bring unique numbers of items, N . Your task is to find and return an integer value representing the sum of all the prime numbers till N . In case, the number of items is 0 or 1, return 0.

Note: Prime numbers are natural numbers that are divisible by only 1 and the number itself.

Input Specification:

input1: An integer value N

Output Specification:

Return an integer value representing the sum of all the prime numbers till N . In case, the number of items is 0 or 1, return 0.

Example 1:

input1 : 10

Output: 17

Explanation:

Here, $N = 10$. The prime numbers between 2 and 10 are 2,3,5,7 and their sum is 17. Hence, 17 is returned as the output.

Example 2:

input1 : 20

Output: 77

Explanation:

Here, $N=20$. The Prime numbers between 2 and 20 are 2,3,5,7,11,19 and their sum is 77 .
Hence, 77 is returned as the output.

Solution



```
def is_prime(num):  
    if num <= 1:  
        return False  
    if num <= 3:  
        return True  
    if num % 2 == 0 or num % 3 == 0:  
        return False  
    i = 5  
    while i * i <= num:  
        if num % i == 0 or num % (i + 2) == 0:  
            return False  
        i += 6  
    return True
```

```
def sum_of_primes(N):  
    if N <= 1:  
        return 0  
    prime_sum = 0  
    for i in range(2, N + 1):  
        if is_prime(i):  
            prime_sum += i  
    return prime_sum  
N=int(input())  
print(sum_of_primes(N))
```

28. Printing Job

There is a printing shop where the customers submit their print jobs at regular intervals, and each job takes a fixed amount of time to complete. You are given an integer value N denoting the total number of print jobs, and X denoting the time duration after which the next print job arrives. Your task is to find and return an integer value representing how long the last job will have to wait in the queue considering one print will take exact 10 minutes.

Input Specification:

input1 : An integer value N representing the number of printing jobs.

input2 : An integer value X representing the time duration after which the next print job arrives.

Output Specification:

Return an integer value representing how long the last job will have to wait in the queue. If in case, there is no waiting time then return 0.

Example 1:

input1 : 4

input2 : 5

Output : 15

Explanation:

Here, $N = 4$ and $X = 5$. Time to complete one print job is 10 mins. So, the time needed for three prints is $3 \times 10 = 30$. The last print job arrives at 15 minutes (3×5). So, the waiting time is $30 - 15 = 15$. Hence, 15 is returned as the output.

Solution

```
def w_time(n, x):  
    if n <= 0 or x <= 0:  
        return 0  
    t_time = (n - 1) * 10  
    l_arrival = (n - 1) * x  
    wait = t_time - l_arrival  
    return max(wait, 0)
```

```
N = int(input())  
X = int(input())  
print(w_time(N, X))
```

29. Product of an Array

You are given an integer array A of size N. Your task is to find and return an integer value representing the product of all the elements in the array.

Input Specification:

input1 : An integer array A

input2 : An integer value N representing the size of the array

Output Specification:

Return an integer value representing the product of all the elements in the array.

Example 1:

input1: {2,4,6,8}

input2:4

Output:384

Explanation:

Here the array is {2,4,6,8} and when all the elements are multiplied by each other ($2*4*6*8=384$) the product obtained is 384 Hence, 384 is returned as the output

Example 2:

input1 : {1,2,3,4,5}

input2 : 5

Output : 120

Explanation:

Here, the array is {1,2,3,4,5} and when all the elements are multiplied by each other ($1 * 2 * 3 * 4 * 5 = 120$) the product obtained is 120. Hence, 120 is returned as the output.

Solution

```
def product_of_array(arr, n):  
    product = 1  
    for num in arr:  
        product *= num  
    return product  
arr1 = list(map(int,input().split()))  
n1 = int(input())  
print(product_of_array(arr1, n1))
```

30. Repeat String

You are given an integer N and a string S. Your task is to find and return a new string which consists of the original string repeated N times.

Input Specification:

input1 : An integer value N

input2 : A string value S

Output Specification:

Return a new string which consists of the original string repeated N times.

Example 1:**input1 :** 3**input2 :** abc**Output :** abcabcabc**Explanation:**

Here, the original string is "abc". When repeated 3 times, we will get "abcabcabc" as the required new string. Hence, abcabcabc is returned as the output.

Example 2:**input1 :** 5**input2 :** xyz**Output :** xyzxyzxyzxyzxyz**Explanation:**

Here, the original string is "xyz". When repeated 5 times, we will get "xyzxyzxyzxyzxyz" as the required new string. Hence, xyzxyzxyzxyzxyz is returned as the output.

Solution

```
def repeat_string(n, s):  
    if n <= 0:  
        return ""  
    return s * n
```

```
N=int(input())  
S=input()  
print(repeat_string(N, S))
```

31. Rock, Paper, Scissors

Two players A and B, are playing the game of Rock, Paper, Scissors. Player A chooses a move represented by a string value M; and the move can be one of the following: 'rock', 'paper', or 'scissors' where,

rock beats scissors

scissors beats paper

paper beats rock

Your task is to find and return a string value representing the winning move for Player B.

Note: The output is case sensitive.

Input Specification:

input1 : A string value M representing the move chosen by Player A.

Output Specification:

Return a string representing the winning move for Player B.

Example 1:

input1 : rock

Output : paper

Explanation:

Here, Player A chooses 'rock'. The winning move for Player B is 'paper', as 'paper' beats 'rock' in the game of Rock, Paper, Scissors. Hence, paper is returned as the output.

Example 2:

input1 : scissors

Solution

```
def winning_move(player_a_move):  
    winning_moves = {  
        'rock': 'paper',  
        'paper': 'scissors',  
        'scissors': 'rock'  
    }  
  
    return winning_moves.get(player_a_move, "")  
M=input()  
print(winning_move(M))
```

Example 1:**input1:**50**input2 :** 200**Output :** 4**Explanation:**

Here, the ship has a capacity of 50 people and needs to transport 200 people. Since the ship can accommodate 50 people at a time, it will need to make 4 rounds to transport all the people from Point A to Point B. In each round, the ship will transport 50 people until all 200 people are transported. Hence, 4 is returned as the output.

32. Ship Rounds

A ship needs to transport a certain number of people from Point A to Point B. The capacity of the ship is denoted by an integer C , and the total number of people to be transported is denoted by an integer N . Your task is to find and return the number of rounds the ship needs to make in order to transport all the people from Point A to Point B.

Note:

One round consists of the ship travelling from Point A to Point B and then returning to Point A.

Input Specification:

input1 : An integer value C representing the capacity of the ship

input2 : An integer value N representing the total number of people to be transported.

Output Specification:

Return an integer value representing the number of rounds the ship needs to make.

Solution



```
def num_rounds(c, n):  
    trips = (n + c - 1) // c  
    return trips
```

```
C=int(input())  
N=int(input())  
print(num_rounds(C, N))
```

33. Snake Water Gun

You are given a string S, which contains the moves played by two players sequentially in a game called Snake Water Gun. The rules of the games are as follows:

Snake beats Water, Water beats Gun and Gun beats Snake.

The move played by player A and the move played by player B in the first round will be stored in the string as "snakewater".

Similarly, in the second round, the moves will be stored in the string as "snakewatergunsake".

There are total N rounds that are played in the game and your task is to find and return an integer value representing how many rounds Player A will win.

Note: The given string is in lowercase.

Input Specification:

input1 : An integer value N, representing the number of rounds.

input2 : A string value S, representing the moves played by the two players.

Output Specification:

Return an integer value representing how many rounds .Player A wins in the game

Example 1:

input1: 3

input2 : watergunwaterwatergunwater

Output : 1

Explanation:

Here we are given that 3 rounds are played by the two players and their moves are "watergunwaterwatergunwater". So, the results of each round are given in following manner:

1. First Round - Player A shows "water" and Player B shows "gun". As water beats gun, player A wins this round.
2. Second Round - Player A shows "water" and Player B shows "water", which results in a draw.
3. Third Round - Player A shows "gun" and Player B shows "water". As water beats gun, player B wins this round.

We can see that Player A wins 1 round, so 1 is returned as the output.

Example 2:

input1 : 2

input2 : snakesnakegun

Output : 0

Explanation:

Here we are given that 2 rounds are played by the two players and their moves are "snakesnakegun". So, the results of each round are given in the following manner:

First Round - Player A shows "snake" and Player B shows "snake", which results in a draw.

Second Round - Player A shows "gun" and Player B shows "gun", which again results in a draw.

We can see that player A does not win any round, so 0 is returned as the output.

Solution



```
def count_wins(n, s):
    beats = {'snake': 'water', 'water': 'gun', 'gun': 'snake'}
    wins = 0
    lengths = {'snake': 5, 'water': 5, 'gun': 3}
    index = 0
    for i in range(n):
        for move in lengths:
            if s[index:index + lengths[move]] == move:
                a_move = move
                index += lengths[move]
                break
        for move in lengths:
            if s[index:index + lengths[move]] == move:
                b_move = move
                index += lengths[move]
                break
        if beats.get(a_move) == b_move:
            wins += 1
    return wins
```

```
N=int(input())
S=input()
print(count_wins(N, S))
```

34. Stabilize the System

In a tech startup, the team faced a software bug - the zeros in their data outputs were causing their system to crash. A junior developer suggested replacing all the 0's with 5's as a quick fix. By implementing this simple code tweak, they stabilized the system. Your task is to find and return an integer value representing the value that stabilizes the system.

Input Specification:

input1 : An integer value

Output Specification:

Return an integer value representing the value that stabilizes the system.

Solution

```
def StabilizeSystem(value):  
    v= str(value)  
    s= v.replace('0', '5')  
    StabilizedValue = int(s)  
    return StabilizedValue
```

```
n=int(input())  
print(StabilizeSystem(n))
```

35. Store Water

You live in a city that is facing a water crisis. The timings when water is supplied in a day are fixed. You have several water tankers that are cylindrical in shape, and know the radius R and height H of the tanker. Your task is to find and return integer value representing the amount (in litres) of water you will be able to ___ particular day.

Note:

Use __pie symbol__ = 3.14

Round of the values to the nearest integer

Input Specification:

input1 : An integer value R representing the radius of the container.

input2 : An integer value H representing the height of the container.

Output Specification:

Return an integer value representing the amount (in litres) of water you are able to store on a particular day.

Solution

```
def WaterStored(R, H):  
    pi = 3.14  
    volume = pi * (R ** 2) * H  
    RoundedValue = round(volume)  
    return RoundedValue  
  
R=int(input())  
H=int(input())  
print(WaterStored(R,H))
```

Example 1:

input1 : 5

input2 : 10

Output : 785**Explanation:**

Here, the given values are $R = 5$ and $H = 10$. The amount of water that can be stored on this day is $3.14 * 5 * 5 * 10 = 785$. Therefore, 785 is returned as the output.

Example 2:

input1 : 7

input2 : 9

Output : 1385**Explanation:**

Here, the given values are $R = 7$ and $H = 9$. The amount of water that can be stored on this day is $3.14 * 7 * 7 * 9 = 1384.7$, which can be rounded up to the value 1385. Therefore, 1385 is returned as the output.

Solution

```
def CountValidSubarrays(arr, N):  
    count = 0  
    for i in range(N - 2):  
        if arr[i] + arr[i + 2] == arr[i + 1]:  
            count += 1  
    return count  
arr=list(map(int,input().split()))  
N=int(input())  
print(CountValidSubarrays(arr,N))
```

Example 1:

input1 :8

Output : 7

Explanation:

The binary representation of 8 is 1000 and the closest symmetrical number with all set bits is 15 with binary representation "1111". Hence, 7 needs to be added to 8 to go make the number symmetrical. Therefore, 7 is returned as the output.

Example 2:

input1 : 2

Output :1

Explanation:

The binary representation of 2 is 10 and the closest symmetrical number with all setbits is 3 with binary representation "11". Hence, 1 needs to be added to 2 to make the number symmetrical. Therefore, 1 is returned as the output.

37. Symmetrical Number

John is an artist who loves everything to be symmetrical. He is given an integer N and he wants to make the number symmetrical in its binary representation which means all the bits are set in the binary form of a number. Your task is to find and return an integer value representing the smallest number which John can add to make the given number symmetrical.

Note: Set bit means 1 in the binary form.

Input Specification:

input1 : An integer value N

Output Specification:

Return an integer value representing the smallest number which John can add to make the given number symmetrical.

Solution



```
def smallest_addition(n):  
    num_bits = n.bit_length()  
    symmetrical_num = pow(2, num_bits) - 1  
    addition = symmetrical_num - n  
    return addition
```

```
N = int(input())  
print(smallest_addition(N))
```

Solution

```
def TallPlantsCount(heights):  
    AverageHeight= sum(heights) / len(heights)  
    RoundedValue= round(AverageHeight)  
    Count= sum(1 for h in heights if h >RoundedValue)  
    return Count  
  
H=list(map(int,input().split()))  
print(TallPlantsCount(H))
```

38. Tall Plants

You are given an array of integers where each element represents the heights of the plants in a garden. A plant is considered "tall" if it is strictly greater than the average height of all plants in the garden. Your task is to find and return an integer value representing the count of tall plants in the garden.

Note: Round off the average to nearest whole number.

Input Specification:

input1 : An integer array representing the height of plants.

Output Specification:

Return an integer value representing the count of tall plants in the garden

Example 1:

input1 : (2,4,5,7,8,14)

input2 : 6

Output : 98

Explanation:

Here, the baked items are 2,4,5,7,8 and 14. 7 and 14 are the only items that are divisible by 7, so their product will be 98. Hence, 98 is returned as the output.

Example 2:

input1 : {11,10,8}

input2 : 3

Output : 0

Explanation: Here, the baked items are 11,10 and 8. Since there is no item that is divisible by 7, 0 is returned as the output

Solution

```
def products(B, N):  
    p= 1  
    d= False  
    for BakedItem in B:  
        if BakedItem % 7 == 0:  
            p*= BakedItem  
            d= True  
    if not d:  
        return 0  
    return p  
  
B=list(map(int,input().split()))  
N=int(input())  
print(products(B,N))
```

40. The Clock Problem

Your friend has problem a reading the time if a clock follows 24-hour format, so you decide to help him out. You have two integers X and Y. Your task is to find and return an integer value representing the product of these two integers in the 12-hour system.

Input Specification:

input1 : An integer value X

input2 : An integer value Y

Output Specification:

Return an integer value representing the product of the two integers in the 12-hour system.

Example 1:

input1 : 4

input2 : 5

Output : 8

Explanation: Here, $X = 4$ and $Y = 5$, and their product is $4 \times 5 = 20$. Therefore, 8 is returned as the output in 12-hour system.

Solution

```
def Product12HourFormat(X, Y):  
    product =X * Y  
    result = product % 12  
    if result == 0:  
        result = 12  
    return result
```

```
X=int(input())  
Y=int(input())  
print(Product12HourFormat(X,Y))
```

Example 2:

input1 : 2

input2 : 5

Output : 10

Explanation:

Here, $X = 2$ and $Y = 5$, and their product is $2*5=10$. Therefore, 10 is returned as the output in 12-hour system.

41. The lost Digit

You are working on a number sequence puzzle. You have an array, A, that is supposed to contain all the numbers from 1 to N. but you realize one number is missing. The array might have been shuffled, so the numbers are not in order. Your task is to find and return an integer value representing the missing number from the sequence,

Input Specification:

input1 : An integer array A

input2 : An integer value N representing the length of the sequence.

Output Specification:

Return an integer value representing the missing number from the sequence.

Example 1:

input1 : [3,1,2,5]

input2 : 5

Output : 4

Explanation:

Here, $A = [3,1,2,5]$ and $N = 5$. The sequence should contain the numbers from 1 to 5 but the number 4 is missing from the array A. Hence, 4 is returned as the output.

Example 2:

input1 : (2,3,4)

input2 : 4

Output : 1

Explanation:

Here, $A = (2,3,4)$ and $N = 4$. The sequence should contain the numbers from 1 to 4 but the number 1 is missing from array A. Hence, 1 is returned as the output.

42. The Maze



In a mobile game, your character has two keys K1 and K2 represented by strings. The character has to navigate through a maze field and upon reaching the end he has to open a gate to finish the game. The gate is locked and can only be opened when the character inputs a passcode. The passcode is the count of substrings common between the two keys. Your task is to find and return an integer value representing the count of common substrings

Note: A single substring for both the keys can only be counted once.

Input Specification:

input1 : A string value K1, representing the first key.

input2 : A string value K2, representing the second key.

Output Specification:

Return an integer value representing the count of common substrings.

Example 1:

input1 : abcdxyz

input2 : xyzabcd

Solution

```
def LostDigit(A,N):  
    ExpectedSum = N * (N + 1) // 2  
    ActualSum= sum(A)  
    MissingDigit=ExpectedSum-ActualSum  
    return MissingDigit
```

```
A=list(map(int,input().split()))  
N=int(input())  
print(LostDigit(A,N))
```

Example 1:

input1 : [3,1,2,5]

input2 : 5

Output : 4

Explanation:

Here, $A = [3,1,2,5]$ and $N = 5$. The sequence should contain the numbers from 1 to 5 but the number 4 is missing from the array A. Hence, 4 is returned as the output.

Example 2:

input1 : (2,3,4)

input2 : 4

Output : 1

Explanation:

Here, $A = (2,3,4)$ and $N = 4$. The sequence should contain the numbers from 1 to 4 but the number 1 is missing from array A. Hence, 1 is returned as the output.

Output : 16

Explanation:

Here, the given string K, is "abcdxyz" and K2 is "xyzabcd". Now, all the common substrings are "a", "ab", "abc", "abcd", "b", "bc", "bcd", "c", "cd", "d", "x", "xy", "xyz", "y", "yz", "z", so the count of common substrings is 16. Therefore, 16 is returned as the output.

Example 2:

input1 : daman

input2 : aman

Output : 9

Explanation:

Here, the given string K1 is "daman" and K2 is "aman". Now, all the common substrings are "a", "am", "ama", "aman", "m", "ma", "man", "an", "n", so the count of common substrings is 9. Therefore, 9 is returned as the output.

Solution

```
def count_common_subs(k1, k2):  
    def get_subs(s):  
        subs = set()  
        l = len(s)  
        for i in range(l):  
            for j in range(i + 1, l + 1):  
                subs.add(s[i:j])  
        return subs  
    subs_k1 = get_subs(k1)  
    subs_k2 = get_subs(k2)  
    common_subs = subs_k1.intersection(subs_k2)  
    return len(common_subs)
```

```
K1 = input().strip()  
K2 = input().strip()  
print(count_common_subs(K1, K2))
```

43. Three Rats

There are three rats in a house where there is a magic cupboard that increases the size of two of the rats and decreases the size of the third rat by 1 unit in a single day. You are given the size of rats as X , Y and Z and your task is to find and return an integer value representing the minimum number of days it will take to make the sizes of all the rats equal. If it is not possible, return -1.

Input Specification:

input1 : An integer value X , representing the size of the first rat

input2 : An integer value Y , representing the size of the second rat

input3 : An integer value Z representing the size of the third rat.

Output Specification:

Return an integer value representing the minimum number of days it will take to make all the rats equal. If it is not possible, return -1.

Example 1:

input1: 3

input2 : 7

input3 : 3

Solution

```
def MinDaysRequired(X, Y, Z):  
    if (X + Y + Z) % 3 != 0:  
        return -1
```

```
    max_size = max(X, Y, Z)
```

```
    days = 0
```

```
    while X != Y or Y != Z:
```

```
        sizes = sorted([X, Y, Z])
```

```
        print(sizes)
```

```
        X, Y, Z = sizes[0], sizes[1], sizes[2]
```

```
        X += 1
```

```
        Y += 1
```

```
        Z -= 1
```

```
        days += 1
```

```
    return days
```

```
X = int(input())
```

```
Y = int(input())
```

```
Z = int(input())
```

```
print(MinDaysRequired(X, Y, Z))
```

44. Unsold Products



In a coding project you are developing a feature for a data visualization tool that processes large datasets of products stored in array A. One dataset contains sales figures for various products, where a sale value of 0 indicates that the product was not sold during a specific period. Your task is to find and return an integer array representing the dataset representing all the unsold products at the end.

Input Specification:

input1 : An integer array A containing sales number of a particular product

Output Specification:

Return an integer array representing the dataset representing all the unsold products at the end.

Example 1:

input1 : [5,2,0,8,0,2,1]

Output : [5,2,8,2,1,0,0]

Solution



```
def unsold_products(arr):  
    sold = [x for x in arr if x != 0]  
    unsold = [x for x in arr if x == 0]  
    return sold + unsold
```

```
A = list(map(int, input().split()))  
print(unsold_products(A))
```

45. Vacant Rooms



You are the manager of a hotel you have a list of rooms already reserved in an array A of size N . You also have the total number of rooms T . Your task is to find and return an integer value representing the total number of vacant rooms.

Input Specification:

input1 : An integer array A representing the reserved rooms.

input2 : An integer value N representing the count of reserved rooms

Input3 : An integer value T representing the total number of rooms.

Output Specification:

Return an integer value representing the total number of vacant rooms.

Example 1:

input1 : [1,2,3,4,5,6,7,8]

input2: 8

input3: 10

Output : 2

Explanation: Here, the given number of total rooms are 10 and the list of occupied rooms have a total of 8 rooms. Therefore, there are a total of 2 ($10 - 8$) rooms which are vacant. Hence, 2 is returned as the output.

Example 2:

input1 : (101,201,301,401)

input2: 4

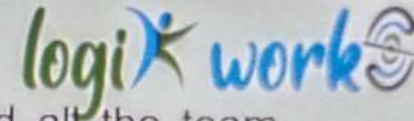
Input3: 15

Output: 11

Solution

```
def VacantRooms(ReservedRooms,ReservedCount,TotalRooms):  
    return TotalRooms-ReservedCount  
  
ReservedRooms =list(map(int,input().split()))  
ReservedCount=int(input())  
TotalRooms=int(input())  
print(VacantRooms(ReservedRooms,ReservedCount,TotalRooms))
```

46. Varsity Print



You are the president of a club in your college. Your annual fest is coming up and all the team members need a varsity shirt with the initials of their name printed on the back. You have the name list of all the members as a string array S . Your task is to find and return a string value representing the initials of each team member.

Input Specification;

input1 : A string array S representing the names of the team members

input2 : An integer value representing the size of the team

Output Specification:

Return a string value representing the initials of each team member.

Example 1:

input1 : (Sam Harris)

input2 : 1

Output : SH

Solution

```
def Initials(names, size):  
    initials = []  
    for n in names:  
        parts = n.split()  
        initial = ".join([part[0].upper() for part in parts])  
        initials.append(initial)  
    return ".join(initials)
```

```
names=input().split()  
size=int(input())  
print(Initials(names, size))
```

47. In a concert hall, there are certain N number of seats arranged in a single row stored in an array where 0 indicates an empty seat and 1 indicates obstruction. You need to accommodate a group of fans. Your task is to find and return an integer value representing the maximum number of fans that can be seated.

Input

input1: {0, 1, 0, 0, 1, 0, 0}

input2: 7

Output: 5

Explanation: Here, the given row is {0, 1, 0, 0, 1, 0, 0}. We can make 1 person seated at index 0. After the obstruction at index 1, there are two available seats. We can use both seats, and after this we can make 5 fans sit in this row. Therefore, 5 is returned as the output.

Input Specification:

input1: An integer array A representing the seating row.

input2: An integer value N representing the number of seats in a row.

Output Specification: Return an integer value representing the maximum number of fans that can be seated.

Solution

```
def maxfSeated(A, N):  
    maxf = 0  
    eSeats = 0  
    for i in A:  
        if i == 0:  
            eSeats += 1  
        else:  
            maxf += eSeats  
            eSeats = 0  
    maxf += eSeats  
    return maxf
```

```
A = list(map(int,input().split()))  
N = int(input())  
print(maxfSeated(A, N))
```

48. Alex gives you a positive number N and wants you to re-arrange the bits of the number in its binary representation such that all the set bits are in consecutive order. Your task is to find and return an integer value representing the minimum possible number that can be formed after re-arranging the bits of the number N .

Note: A set bit means 1 in the binary form of a number.

Input Specification:

input1: An integer value N

Output Specification: Return an integer value representing the minimum possible number that can be formed after re-arranging the bits of the number N .

Example 1:

Input

input1: 10

Output: 3

Explanation: Here, the given number is 10, whose binary representation is 1010. So we can rearrange the bits to make all the set bits consecutive, as 0011, which is the lowest possible number a set bits consecutive combinations. Therefore, 3 is returned as the output.

49. You are on a hiking trail represented by an array A of length N, where the trail initially ascends and then descends, forming a single peak. Your task is to find and return an integer value representing the elevation of the summit.

Input Specification:

input1: An integer array A

input2: An integer value N denoting the size of A

Output Specification: Return an integer value representing the elevation of the summit.

Input

input1: {1, 2, 3, 4, 3, 2, 1}

input2: 7

Output: 4

Explanation: Here, in the array {1, 2, 3, 4, 3, 2, 1} we can see that the values are increasing till 4 and then the value starts to decrease, therefore, 4 will be the peak. Hence 4 is returned as the output.

Solution

```
def peakEle(A, N):  
    for i in range(1, N - 1):  
        if A[i] > A[i - 1] and A[i] > A[i + 1]:  
            return A[i]  
    return -1
```

```
A = list(map(int, input().split()))  
N = int(input())  
print(peakEle(A, N))
```

50. Bob goes to supermarket to shop candies represented by an array A for Halloween party, his mother gave him some money M . Due to the festive season, there are several offers in the supermarket. One such offer useful for Bob is, if the price of the candy is a multiple of 5, he can buy it without spending any money otherwise he will spend money equal to A_i which is the price of a particular candy. Bob can shop as long as he has money. Your task is to find and return the maximum number of candies Bob can buy.

Note: Assume 1 - based indexing.

Input Specification:

input1: An integer value, representing number of candies

input2: An integer array A , representing price of candy

input3: An integer value M , representing the amount of money.

Example 1:

Input

input1: 3

input2: {5, 5, 105}

Output: 16

Output Specification: Return the number of candies Bob can buy.

Solution

```
def maxCandies(N, A, M):  
    count = 0  
    for i in A:  
        if i % 5 == 0:  
            count += 1  
        elif M >= i:  
            M -= i  
            count += 1  
        else:  
            break  
    return count
```

```
N = int(input())  
A = list(map(int, input().split()))  
M = int(input())  
print(maxCandies(N, A, M))
```

51. Noah is given an integer array A of length N . He must perform the following operations on the array:

Select any integer pair/s from the array with their sum equal to 18.

From this select the pair with the maximum product such that the first element of the pair is greater than the second element of the pair.

Your task is to help Noah find and return a pair in the form of an integer array which satisfies the conditions mentioned.

Input Specification:

input1: An integer value N , representing the size of array A

input2: An integer array A

Output Specification:

Return a pair in the form of an integer array which satisfies the conditions mentioned.

Input

input1: 8

input2: {11, 1, 2, 8, 10, 11, 15, 7}

Output: {10, 8}

Explanation

The given array is {11, 1, 2, 8, 10, 11, 15, 7}

There are four pairs in the array, {10, 8}, {8, 10}, {11, 7} and {7, 11} with their sum equal to 18.

Pair {8, 10} and {7, 11} are not valid as the first element in each pair is less than the second element in the pair.

The product of the pair {10, 8} is 80 and that of {11, 7} is 77.

Since the pair {10, 8} has a higher product, {10, 8} is returned as the output.

52. Charles is given an array A. He wants to find the count of occurrence of second largest element in the array. Your task is to help him find and return an integer value representing the count of occurrence of the second largest element in an array.

Note: If the array contains the same elements, then return 0. The array has only consecutive elements

Input Specification:

input1: An integer N, representing length of array.

input2: An integer array A

Output Specification: Return an integer value representing the count of occurrence of the second largest element in an array.

Input

input1: 8

input2: {1, 2, 3, 4, 4, 5, 5, 5}

Output

2

Explanation: Here the length of array A is 8(N).

The second largest element is 4 and the occurrence of 4 is 2 times. Therefore, 2 is returned as the output.

Solution

```
def maxProPair(N, A):
    max_product = -1
    result_pair = []
    for i in range(N):
        for j in range(i + 1, N):
            if A[i] + A[j] == 18:
                if A[i] > A[j]:
                    product = A[i] * A[j]
                    if product > max_product:
                        max_product = product
                        result_pair = [A[i], A[j]]
                elif A[j] > A[i]:
                    product = A[j] * A[i]
                    if product > max_product:
                        max_product = product
                        result_pair = [A[j], A[i]]
    return result_pair

N = int(input())
A = list(map(int, input().split()))
print(maxProPair(N, A))
```

Solution

```
def SecLargest(N, A):  
    largest = max(A)  
    second_largest = float('-inf')  
    for num in A:  
        if num != largest and num > second_largest:  
            second_largest = num  
    if second_largest == float('-inf'):  
        return 0  
    count = A.count(second_largest)  
    return count
```

```
N = int(input())  
A = list(map(int, input().split()))  
print(SecLargest(N, A))
```

Solution

```
def UniquePairs(N, A):  
    count = 0  
    pairs = set()  
    for i in range(N):  
        for j in range(i + 1, N):  
            if (A[i] * A[j]) % 3 == 0:  
                pair = tuple(sorted((A[i], A[j])))  
                if pair not in pairs:  
                    pairs.add(pair)  
                    count += 1  
    return count  
  
N = int(input())  
A = list(map(int, input().split()))  
print(UniquePairs(N, A))
```

54. You are stuck on an island where they sell and eat coconut sweets only.

A person can buy at most 1 box per day with each box containing N pieces. To remain alive, you must consume E coconut sweets daily for D days, but the catch is that you cannot purchase sweets on Sundays. Your task is to find and return an integer value representing the minimum number of times you have to buy coconut sweets in order to stay alive. If not possible, return -1.

Note: The day starts from Monday

Input Specification:

input1: An integer value N , representing the number of coconut sweets per box.

input2: An integer value E , representing the number of coconut sweets you must eat daily

input3: An integer value D , representing the number of days you have to spend on the island

Output Specification: Return an integer value representing the minimum number of times you have to buy coconut sweets in order to stay alive. If not possible, return -1.

Solution

```
def MinBoxes(N, E, D):  
    total = E * D  
    count = D // 7  
    rem = D % 7  
    pdays = count * 6 + min(rem, 6)  
    if pdays * N < total:  
        return -1  
    min_boxes = (total + N - 1) // N  
    return min_boxes
```

N = 16

E = 2

D = 10

```
print(MinBoxes(N, E, D))
```

Example 1:**Input:**

input1: 16

input2: 2

input3: 10

Output: 2

Explanation: Here, there are 16 coconut sweets per box. You have to live for 10 days on the island and each day you will have to eat 2 sweets. So in 8 days you will run out of sweets and buy another box on the 9th day.

Now, this second box will last for the next 8 days, which means that you can eat till 16th day but you will have to leave the island after 10th day, so you don't need to buy any more boxes. since you have bought sweet boxes 2 times, 2 is returned as the output.

55. You are developing a feature for an environmental awareness app that helps users to know how much area their tree's shadow covers. You have the distance D from a tree's trunk to the edge of the shadow. Your task is to calculate and return an integer value representing the shadow area of the canopy

Note:- Round off the result to nearest integer.

Input Specification: input1: An integer value D , representing the distance from the tree trunks to the edge of shadow.

Output Specification: Return an integer value representing the shadow area of the canopy.

Input :5

Output: 79

Explanation: Here, $D = 5$. So, area of the canopy would returned as output.

I be $3.14 * 5 * 5 = 78.5$. Therefore, 79 will be returned as output

Solution

```
import math
```

```
def shadow_area(D):  
    area = math.pi * D * D  
    return math.floor(area)
```

```
D = int(input())  
print(shadow_area(D))
```

56. You are given a string S and your task is to find and return the count of permutations formed by fixing the positions of the vowels present in the string.

Note: Ensure the result is non negative. If there are no consonants then return 0.

Input Specification:

input1: A string S

Output Specification: Return an integer value representing the count of permutations formed by fixing the positions of the vowels present in the string.

Example 1:

input1: ABC

Output: 2

Explanation: Here, in the given string "ABC" there is 1 vowel ('A') and by fixing its position there are 2 permutable letters ('B', 'C'). So, its permutation is $2! = 2$. Therefore, 2 is returned as the output.

57. You are given a string S of length N . Your friend wants to know the number of times his favourite letter C occurs in the string. Your task is to help your friend find and return an integer value representing the number of times a character occurs in a particular string. Note: All the characters in the strings are in lowercase.

Input Specification:

input1: A string S

input2: An integer N , representing the length of string

input3: A character C

Output Specification: Return an integer value representing the number of times a character occurs in a particular string.

input1: helloworld

Input2: 10

input3: l

Output: 3

Explanation: Here the given string is "helloworld" and the length of the string is 10 and the character whose count is to be found is "l". The number of times "l" occurs in the string is 3. Therefore, 3 is returned as the output.

Solution

```
import math
def count_permutations(s):
    vowels = set('aeiouAEIOU')
    consonants_count = 0
    for char in s:
        if char not in vowels:
            consonants_count += 1
    if consonants_count == 0:
        return 0
    return math.factorial(consonants_count)

s = input()
print(count_permutations(s))
```

Solution

```
def count_char(s, n, c):  
    return s.count(c)
```

```
s = input()  
n = int(input())  
c = input()  
print(count_char(s, n, c))
```

input1: thunder

input2: (puzzle,thunder,powder, blender,under}

input3: 5

Output: under

Explanation: Here, the given word S is "thunder." We can find the rhyming word/s from the dictionary in the following way:

The first word "puzzle" does not rhyme at all and has no common characters with the word "thunder" at the end.

The second word "thunder" is exactly the same word as the given word, so it will not be considered as a rhyming word.

The third word is "powder" and has the 3 letters "der" in common at the end and is the least rhyming among the list.

The fourth word is "blender" and has the 4 letters "nder" in common at the end and is the second most rhyming among the list.

The last word is "under" and it is the most rhyming word as it has the 5 letters "under" in common with the word at the end.

Since "under" is the most rhyming word among those in the given dictionary, under is returned as the output.

Solution



```
def rhyme_best(s, d, l):  
    def suffix_len(w1, w2):  
        max_len = min(len(w1), len(w2))  
        for i in range(max_len):  
            if w1[-1 - i:] == w2[-1 - i]:  
                continue  
            return i  
        return max_len  
    best = "No Word"  
    max_len = 0  
    for w in d:  
        if w != s:  
            cur_len = suffix_len(s, w)  
            if cur_len > max_len:  
                max_len = cur_len  
                best = w  
    return best  
  
s = input()  
d = list(map(str, input().split()))  
l = int(input())  
print(rhyme_best(s, d, l))
```

59. Encode the message



You are a secret agent on a mission, and need to decode various messages using the ASCII values of characters. Due to resource constraints, you can send only those messages with a total ASCII sum equal to or less than a specified threshold N . You need to remove characters from the end of the string to maintain the sum accordingly. Your task is to find and return the decoded string, or return an empty string, if it is not possible to meet the condition.

Input Specification:

input1: A string value S , representing the encoded message

input2: An integer value N , representing the threshold value

Output Specification:

Return a string value representing the decoded string, or an empty string if it is not possible to meet the condition.

Example 1:

Input1: AlohoMora

Input2: 584

Output: AlohoM



Explanation:

Here, the given string is "AlohoMora". The ASCII sum of the characters is $65 + 108 + 111 + 104 + 111 + 77 + 111 + 114 + 97 = 898$. Now we will remove the characters in the following manner.

We will first remove the character 'a' and the sum will become $898 - 97 = 801$.

Next we will remove the character 'r' and the sum will become $801 - 114 = 687$.

Then we will remove the character 'o' and the sum will become $687 - 111 = 576$, which is less than 584.

Therefore, AlohoM is returned as the output.

Solution

```
def decode_message(S, N):  
    total_sum = sum(ord(c) for c in S)  
    if total_sum <= N:  
        return S  
    for i in range(len(S) - 1, -1, -1):  
        total_sum -= ord(S[i])  
        if total_sum <= N:  
            return S[:i]  
    return ""
```

```
S = input()  
N = int(input())  
print(decode_message(S, N))
```

60. Logical Operation:

You are given two integers A and B. Your task is to find and return an integer representing the value of their logical OR operation.

Note: Logical OR is a binary operator returning true if at least one operand is true; false if both are false i.e.

$$1 \text{ OR } 0 = 1$$

$$0 \text{ OR } 1 = 1$$

$$0 \text{ OR } 0 = 0$$

$$1 \text{ OR } 1 = 1$$

Input Specification:

input1: An integer value A

input2: An integer value B

Output Specification: Return an integer representing the value of their logical OR operation:

Example 1:

Input1: 5

Input2: 9

Output: 5

Solution

```
def logical_or(A, B):  
    if A or B:  
        return 1  
    return 0  
  
A = int(input())  
B = int(input())  
print(logical_or(A, B))
```

61. Magical Number



You are given a program to find the count of magical numbers from 1 to N. A magical number is defined by the following criteria:

Convert each number in the range 1 to N (inclusive) to its binary representation. Replace '0' with '1' and '1' with '2' in the binary string. Calculate the sum of the digits in the modified binary string. If the resultant number is odd, then it is considered a magical number.

Your task is to find and return an integer value representing the count of the magical numbers present within the given range.

Input Specification:

input1: An integer value N representing the range of numbers.

Output Specification: Return an integer value representing the count of magical numbers present within the range.

Example 1:

input1: 2

output: 1

Explanation:

For the input ($N = 2$), let's go through the steps to find the count of magical numbers:

1. Number 1:
 - Binary representation: 1
 - Modified binary: 2 (replace '1' with '2')
 - Sum of digits: 2 (even, not magical)
2. Number 2:
 - Binary representation: 10
 - Modified binary: 21 (replace '1' with '2' and '0' with '1')
 - Sum of digits: $2 + 1 = 3$ (odd, magical)

So, there is 1 magical number in the range from 1 to 2.

Solution

```
def count_magical_nums(N):  
    count = 0  
    for num in range(1, N + 1):  
        bin_rep = bin(num)[2:]  
        mod_bin_list = list(bin_rep)  
        for i in range(len(mod_bin_list)):  
            if mod_bin_list[i] == '0':  
                mod_bin_list[i] = '1'  
            elif mod_bin_list[i] == '1':  
                mod_bin_list[i] = '2'  
        mod_bin = ''.join(mod_bin_list)  
        digit_sum = sum(int(digit) for digit in mod_bin)  
        if digit_sum % 2 == 1:  
            count += 1  
    return count  
  
N = int(input())  
print(count_magical_nums(N))
```

62. Minimum Badness



You are given a string S representing the colours of houses (red(R), blue(B) or white(W)). The badness value is the number of differently-coloured adjacent houses. You can paint the white houses red or blue or minimize the badness value. Your task is to find and return an integer value representing the minimum possible badness value.

Input Specification: A string value S , representing the colours of houses.

Output Specification: Return an integer value representing the minimum possible badness value.

Example 1:

Input: RRWBWBW

Output: 1

Explanation: The output is 1 because you can paint all the white houses (W) to match the color of the adjacent houses, resulting in a single color change. For example, painting the string "RRWBWBW" to "RRRBRBR" minimizes the badness value to 1.

Solution

```
def minBadness(S):  
    h = list(S)  
    n = len(h)  
    for i in range(n):  
        if h[i] == 'W':  
            if i > 0 and h[i - 1] != 'W':  
                h[i] = h[i - 1]  
            elif i < n - 1 and h[i + 1] != 'W':  
                h[i] = h[i + 1]  
    b = 0  
    for i in range(1, n):  
        if h[i] != h[i - 1]:  
            b += 1  
    return b  
  
S = input()  
print(minBadness(S))
```

Example 1:

input1:18

Output: 162

Explanation: Here the given initial energy is 18. Now you can advance to higher levels in the following manner:

The initial energy that you have is 18 and the energy used to advance to the next level is 16 (which is the highest power of 2 less than or equal to 18). Now the updated energy is 2 ($18 - 16$) and the score becomes 16.

The updated energy is 2, and to advance to the next level, the energy used is 2. Now, the updated energy level is 2 and the score becomes 162.

The energy is now completely utilised, and the maximum score achieved is 162. Therefore 162 is returned as the output.

Example 2:

Input1: 2

Output: 2

Explanation: For $N = 2$, the highest power of 2 less than or equal to 2 is 2, so $N = 0$ and final score is 2. Since $N = 0$, we now have to stop. Therefore, 2 is returned as the output.

63. Maximum Levels



You are playing a game and want to cross as many levels as possible in the game. To advance from one level to another, these are the rules:

- A player has an initial energy **N** from which you have to spend the highest power of 2, i.e, **X**, from the current level in order to go to the next level.
- The value of X is then concatenated to the score for each level you cross.

Your task is to find and return an integer value representing the final concatenated score by crossing the maximum possible levels with the given initial energy.

Input Specification: An integer value N, representing the initial energy.

Output Specification: Return an integer value representing the final concatenated score by crossing the maximum possible levels with the given initial energy.

64. Odd Sum



You are the class teacher of **N** students, and are 19 assigning them a task in which they need to find the odd parts of all their roll numbers. The odd parts of the roll numbers can be calculated using the following formula:

- $r[i] = (2^k) * t$ (where r is the roll number of the i^{th} student, k is the nearest maximum power of 2 less than i^{th} number and t is the odd part of the roll number)

Your task is to find and return an integer value representing the sum of the odd parts of the roll numbers of every student in the class.

Note: Return the output module 10007

Input Specification: An integer value N , representing the

Output Specification: Return an integer value representing the sum of the odd part of the roll numbers of every student in the class

Example 1:**input1:6****Output: 14****Explanation:**

Here, the number of students in the class is 6. So, we can find the odd parts of their roll numbers as follows:

1. Roll no. 1 = $(2^0)*1$, so t is 1.
2. Roll no. 2 = $(2^1)*1$, so t is 1.
3. Roll no. 3 = $(2^0)*3$, so t is 3.
4. Roll no. 4 = $(2^2)*1$, so t is 1.
5. Roll no. 5 = $(2^0)*5$, so t is 5.
6. Roll no. 6 = $(2^1)*3$, so t is 3.

As we can see, the sum of the odd parts of their roll numbers = $1 + 1 + 3 + 1 + 5 + 3 = 14$
Therefore, **14** is returned as the output.

Example 2:**input1:2****Output: 2****Explanation:**

Here, the number of students in the class is 6. So, we can find the odd parts of their roll numbers as follows:

1. Roll no. 1 = $(2^0)*1$, so t is 1.
2. Roll no. 2 = $(2^1)*1$, so t is 1.

As we can see, the sum of the odd parts of their roll numbers = $1 + 1 = 2$. Therefore, **2** is returned as the output.

65. Diwali Contest



Max is planning to take part in a Diwali contest at a Diwali Party that will begin at 8 PM and will run until midnight (12 AM) i.e. for 4 hours. He also needs to travel to the party venue within this time which takes him P minutes. The contest comprises of N problems that are arranged in order of difficulty, with problem 1 being the simplest and problem N being the most difficult. Max is aware that he will require $5 * i$ minutes to solve the i th problem.

Your task is to help Max find and return an integer value, representing the number of problems Max can solve and reach the party venue within the given time frame of 4 hours.

Note: Max will leave his home at exactly 8 PM to reach the party venue.

Input Specification:

Input1: An integer value N , representing the total number of problems.

Input2: An integer value P , representing the time to travel from his home to the party venue.

Output Specification: Return an integer value, representing the number of problems Max can solve and reach the party venue within the given time frame of 4 hours.

Solution

```
def odd_part_sum(n):  
    total_sum = 0  
    for i in range(1, n + 1):  
        num = i  
        while num % 2 == 0:  
            num //= 2  
        total_sum += num  
    return total_sum % 10007
```

```
n = int(input())  
print(odd_part_sum(n))
```

Example 1:

Input1: 4

Input2: 191

Output: 3

Explanation: Here Max has a total of 4 problems, and it will take him 191 minutes to reach the party venue. We can find the number of problems that Max can solve by the following way:

The total time Max has is 4 hours, i.e. 240 minutes.

Max needs 191 minutes to reach the party venue.

Time left to solve the problems is 49 minutes.

Max will be able to solve 3 problems which will take him 30 minutes.

Since in the time left, no new problem can be solved. Hence 3 is returned as the output.

Example 2:

Input1: 8

Input2: 30

Output: 8

Explanation: Here Max has a total of 8 problems and it will take him 30 minutes to reach the party venue. We can find the number problems that Max can solve by the following way:

The total time Max has is 4 hours i.s 240 minutes

Max needs 30 minutes to reach the party venue.

Time left to solve the problems is 210 minutes.

Max will be able to solve all 8 problems which will take him 180 minutes.

Hence, 8 is returned as the output.



Solution

```
def max_problems_solved(N, P):  
    total_time = 240  
    time_for_solving = total_time - P  
    if time_for_solving <= 0:  
        return 0  
    time_spent = 0  
    problems_solved = 0  
    for i in range(1, N + 1):  
        time_needed = 5 * i  
        time_spent += time_needed  
        if time_spent > time_for_solving:  
            break  
        problems_solved += 1  
    return problems_solved  
  
N = int(input())  
P = int(input())  
print(max_problems_solved(N, P))
```

Solution

```
def dog_years_to_human_years(N):  
    return N * 7
```

```
N =int(input())  
print(dog_years_to_human_years(N))
```

66. Dog Age

Max has a dog, which is an integer N year old. Now he wants the age of his dog in human years. The internet says that 1 human year equals to 7 dog years. Your task is to find and return an integer value representing the age of Max's dog in human years.

Input Specification: An integer value N representing the age of Max's dog.

Output Specification: Return an integer value representing the age of Max's dog in human years.

Example 1:

Input1: 1

Output: 7

Explanation: Here, the given value of $N = 1$. The age of dog is 1 dog year which is equivalent to 7 in human years. Therefore, 7 is returned as the output.

Example 2:

Input1: 4

Output: 28

Explanation:

Here, the given value of $N = 4$. The age of dog is 4 dog year which is equivalent to 28 in human years. Therefore, 28 is returned as the output

67. Unique Files

You are given a string array S of length N , which contains some unique file names. Your task is to read the first letter of each file and return a string value that denotes the first letter of the files in alphabetical order. In case no element is present, return 0 as a string.

Input Specification:

Input1: An integer value N representing the number of files.

Input2: A string array S representing file names

Output Specification: Return a string value that denotes the first letter of the files in alphabetical order.

Example 1:

Input1: 2

Input2: {name, file}

Output: fn

Explanation: Here, $N = 2$ and file names are {name, file}

name read as "n", file read as "f".

Therefore, fn is returned as the output.

Example 2:

Input:

Input1: 4

Input2: {name, edit surname, edit, editing}

Output:

eeens

68. Leap Year



You are an app developer, and your company wants you to develop a new feature to engage more users by showing the next nearest leap year for a given year, represented by an integer value **N**. Your task is to find and return an integer value representing the next nearest leap year for the given year. If the given year itself is a leap year, then return that.

Input Specification: An integer value N representing the given year.

Output Specification: Return an integer value representing the next nearest leap year for a given year.

Example 1:

Input1:2021

Output: 2024

Explanation: Here, the given year is 2021 and the next upcoming leap year would be 2024. Therefore, **2024** is returned as the output.

Example 2:

input1:2008

Output:2008

Explanation:

Here, the given year is 2008, which is itself a leap year. Therefore **2008** is returned as the output.

Solution

```
def is_leap_year(year):  
    return (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)
```

```
def next_leap_year(N):  
    while not is_leap_year(N):  
        N += 1  
    return N
```

```
N = int(input())  
print(next_leap_year(N))
```

69. Store Water

You live in a city that is facing a water crisis. The timings when water is supply day are fixed. You have several water tankers that are cylindrical in shape, and know the radius R and height of the tanker. Your task is to find and return integer value representing the amount (in litres) of water you will be able to a particular day.

Note:

- Use $\pi = 3.14$
- Round of the values to the nearest integer

Input Specification:

input1: An integer value R representing the radius of the container

input2: An integer value H representing the height of the container.

Output Specification: Return an integer value representing the amount in litres of water you will be able to store on a particular day.

Example 1:

input1:5

input2: 10

Output: 785

Explanation: Here, the given values are $R = 5$ and $H = 10$. The amount of water that can be stored on this day is $3.14 * 5 * 5 * 10 = 785$. Therefore, **785** is returned as the output.

Example 2:

input1:7

input2: 9

Output: 1385

Explanation: Here, the given values are $R = 7$ and $H = 9$. The amount of water that can be stored on this day is $3.14 * 7 * 7 * 9 = 1384.7$, which can be rounded up to the value **1385**. Therefore, 1385 is returned as the output.

70. Olivia's Garden

Olivia has a garden with various types of flowers, and she has been given an array A of size N that represents the beauty scores of each flower. Her mother has instructed her to perform a few operations on the flowers in Order to get the maximum beauty score.

Following are the operations that Olivia needs to perform:

1. First, she must select a specific flower.
2. She must then subtract the selected flower's beauty score from all of the flowers' beauty scores in the array A .
3. Then, she must add the new beauty scores of the flowers and determine the overall score.

Your task is to help Olivia to find and return an integer value, representing the maximum overall beauty score after performing the above operations as instructed by her mother.

Note: The operations can be performed only once.

- Array elements can be negative.
- Assume 0 based indexing.

2nd Way:

Olivia selects the flower at index 0 with beauty score 1.

She will then subtract 1 from all the beauty scores in the array, leading to (0,3,1,8,4). The sum of the beauty scores is $0+3+1+8+4=16$,

Since 16 is the maximum beauty score obtained, 16 is returned as the output.

Example 2:

input1: (2,2,5)

input2: 3

Output: 3

Explanation:

Here in this example, the array given is (2,2,5).

1st way:

Olivia selects the flower at index 0 with beauty score 2.

She will then subtract 2 from all the beauty scores in the array, leading to (0,0,3). The sum of the beauty scores is $0+0+3=3$.

Since 3 is the maximum beauty score obtained, 3 is returned as the output.

Input Specification:

input1: An integer array A, representing the beauty scores of the flowers.

input2: An integer value N, representing size of array A..

Output Specification: Return an integer value, representing the maximum overall beauty score after performing the above operations as instructed by her mother.

Example 1:

input1: (1,4,2,9,5)

input2: 5

Output: 16

Explanation: Here in this example, the array given is (1,4,2,9,5)

1st Way: Olivia selects the flower at index 2 with beauty score 2.

She will then subtract 2 from all the beauty scores in the array, leading to (-1,2,0,7,3). The sum of the beauty scores is $-1+2+0+7+3=11$.